

E. Propagating tree

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Iahub likes trees very much. Recently he discovered an interesting tree named propagating tree. The tree consists of n nodes numbered from 1 to n , each node i having an initial value a_i . The root of the tree is node 1.

This tree has a special property: when a value val is added to a value of node i , the value $-val$ is added to values of all the children of node i . Note that when you add value $-val$ to a child of node i , you also add $-(-val)$ to all children of the child of node i and so on. Look an example explanation to understand better how it works.

This tree supports two types of queries:

- "1 $x\ val$ " — val is added to the value of node x ;
- "2 x " — print the current value of node x .

In order to help Iahub understand the tree better, you must answer m queries of the preceding type.

Input

The first line contains two integers n and m ($1 \leq n, m \leq 200000$). The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 1000$). Each of the next $n-1$ lines contains two integers v_i and u_i ($1 \leq v_i, u_i \leq n$), meaning that there is an edge between nodes v_i and u_i .

Each of the next m lines contains a query in the format described above. It is guaranteed that the following constraints hold for all queries: $1 \leq x \leq n, 1 \leq val \leq 1000$.

Output

For each query of type two (print the value of node x) you must print the answer to the query on a separate line. The queries must be answered in the order given in the input.

Examples

input	Copy
5 5 1 2 1 1 2 1 2 1 3 2 4 2 5 1 2 3 1 1 2 2 1 2 2 2 4	
output	Copy
3 3 0	

Note

The values of the nodes are [1, 2, 1, 1, 2] at the beginning.

Then value 3 is added to node 2. It propagates and value -3 is added to it's sons, node 4 and node 5. Then it cannot propagate any more. So the values of the nodes are [1, 5, 1, -2, -1].

Then value 2 is added to node 1. It propagates and value -2 is added to it's sons, node 2 and node 3. From node 2 it propagates again, adding value 2 to it's sons, node 4 and node 5. Node 3 has no sons, so it cannot propagate from there. The values of the nodes are [3, 3, -1, 0, 1].

You can see all the definitions about the tree at the following link:
[http://en.wikipedia.org/wiki/Tree_\(graph_theory\)](http://en.wikipedia.org/wiki/Tree_(graph_theory))

Codeforces Round 225 (Div. 2)

Finished

Practice



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Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

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Submission	Time	Verdict
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339169888	Sep/17/2025 21:31	Happy New Year!

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data structures

dfs and similar

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trees

*2000

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Codeforces Round #225

Tutorial (en)